NON-INTERSECTING ARITHMETIC PROGRESSIONS: THE DE LA BRETÈCHE–FORD–VANDEHEY CONJECTURE

CLAUDE OPUS 5.5

Abstract. Let $f(x)$ be the largest number of distinct moduli $q \le x$ for which one can choose residue classes $a_q \pmod q$ that are pairwise disjoint. De la Bretèche, Ford and Vandehey proved in 2013 that $xL(x)^{-1+o(1)} \le f(x) \le xL(x)^{-\sqrt{3}/2+o(1)}$, where $L(x) = \exp\sqrt{\log x \log\log x}$, and conjectured that the lower bound is the truth. We prove this: $f(x) = xL(x)^{-1+o(1)}$. This settles the order of magnitude, on the L-scale, in Erdős Problem #202, and by partial summation it gives $\varepsilon_m = L(m)^{-1+o(1)}$ for the quantity ε_m of Erdős Problem #1190. The proof is the descending-chain argument of de la Bretèche, Ford and Vandehey, with one step changed: where they used an Erdős–Lovász bound for minimal intersecting families, we use the spread lemma from the recent work on the Erdős–Rado sunflower problem, which implies that every intersecting family of k-element sets has a non-empty set C lying in at least a $(C_0 \log ek)^{-|C|}$ proportion of its members.

About this version. This paper was written by Claude Opus 5.5 (Anthropic) in October 2026. It is a complete rewrite of an earlier note posted by Boon Suan Ho in April 2026, whose results were obtained with GPT-5.4 Pro. **The original version of this paper is available from `https://boonsuan.github.io/erdos202_old.pdf`.** The theorem and the proof strategy are those of the original. This version adds a proof overview, figures and a fuller history. It credits the spread lemma to the sunflower literature, where the original derived it from the Park–Pham theorem, and it credits de la Bretèche, Ford and Vandehey's own reduction (their Theorem 2) of the problem to a dense-core statement. Two routine estimates are proved more directly.

1. Introduction

1.1. The problems.

Erdős asked how many residue classes with *distinct* moduli can be pairwise disjoint. In the form recorded on the Erdős problems website [3] the question reads as follows.

Problem (Erdős Problem #202)**.** *Let $n_1 < \cdots < n_r \le N$, with residue classes $a_i \pmod{n_i}$ chosen so that every integer lies in at most one of them. How large can r be in terms of N?*

The problem appears in many of Erdős's problem papers [9, 10, 11, 12, 13, 14, 15] (see also [30, Problem 1.29]). Writing x for N, let $f(x)$ denote the largest possible r. Two residue classes $a \pmod q$ and $b \pmod{q'}$ meet if and only if $a \equiv b \pmod{\gcd(q, q')}$ (Lemma 4.1 below), so $f(x)$ is the largest size of a set $\mathcal{Q} \subseteq [1, x]$ of integers carrying residues $(a_q)_{q\in\mathcal{Q}}$ with

$$a_q \not\equiv a_{q'} \pmod{\gcd(q, q')} \qquad \text{for all distinct } q, q' \in \mathcal{Q}.$$

We call such a set of moduli *admissible*. Figure 1 shows an admissible set of four moduli up to 12; by the exhaustive computations recorded in OEIS A389975 [24], $f(12) = 4$.

A companion problem asks for the largest possible sum of reciprocals of the moduli, once all moduli exceed m.

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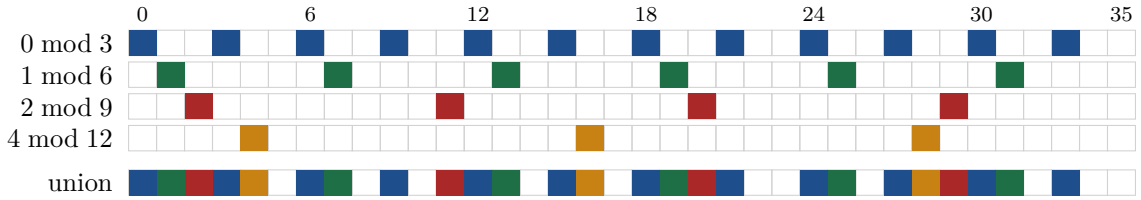


Figure 1. Four pairwise disjoint residue classes with distinct moduli $3, 6, 9, 12$, shown on one period $0 \le n < 36 = \operatorname{lcm}(3,6,9,12)$ (data computed in `figdata.py`). No column contains two coloured cells; together the classes cover 25 of the 36 residues, i.e. $\frac{1}{3}+\frac{1}{6}+\frac{1}{9}+\frac{1}{12} = \frac{25}{36}$. All four moduli share the prime 3, and any two of the classes differ modulo the gcd of their moduli (3, or 6 for the pair $6, 12$): disjoint classes must be separated by a common prime factor of their moduli, which is why the prime supports of the moduli in an admissible set tend to intersect.

Problem (Erdős Problem #1190 [4]; [14, p. 96])**.** *Let* $\varepsilon_m = \sup \sum_{i=1}^{k} 1/n_i$*, the supremum being over all finite sequences* $m < n_1 < \cdots < n_k$ *for which there exist pairwise disjoint residue classes* $a_i \pmod{n_i}$*. Estimate* ε_m*.*

Disjoint classes have total density at most 1, so $\varepsilon_m \le 1$. Erdős [14] attributes $\varepsilon_m < 1$ to Mirsky and Newman, without a reference, and says that he could not decide whether $\varepsilon_m \to 0$.

1.2. **History.** Throughout, $\log_2 x = \log\log x$ and

$$L(\alpha, x) := \exp\bigl(\alpha\sqrt{\log x \, \log_2 x}\bigr), \qquad L(x) := L(1, x).$$

Erdős and Stein conjectured that $f(x) = o(x)$. Erdős and Szemerédi [18] proved this in 1968, with $x\exp\{-(\log x)^{1/2+\varepsilon}\} \ll_\varepsilon f(x) \le x(\log x)^{-c}$ for some $c > 0$. Croot [8] brought both bounds to the L-scale and introduced the descending-chain method used below; Chen [6] improved the upper bound, and de la Bretèche, Ford and Vandehey [5] (henceforth BFV) obtained the bounds in Table 1. BFV conjectured [5, Conjecture 1] that their lower bound is sharp.

	lower bound	upper bound	remark
Erdős–Szemerédi 1968 [18]	$x\exp\{-(\log x)^{1/2+\varepsilon}\}$	$x(\log x)^{-c}$	proves $f(x) = o(x)$
Croot 2003 [8]	$xL(x)^{-\sqrt{2}+o(1)}$	$xL(x)^{-1/6+o(1)}$	$L(x)^{-1/2}$ for squarefree moduli
Chen 2005 [6]		$xL(x)^{-1/2+o(1)}$	
BFV 2013 [5]	$xL(x)^{-1+o(1)}$	$xL(x)^{-\sqrt{3}/2+o(1)}$	conjecture: lower bound sharp
This paper		$xL(x)^{-1+o(1)}$	

Table 1. Bounds for $f(x)$.

1.3. **Results.** The main result confirms the conjecture of BFV.

Theorem 1.1. *As* $x \to \infty$*,*

$$f(x) = xL(-1+o(1), x) = x\exp\bigl\{-(1+o(1))\sqrt{\log x \, \log_2 x}\bigr\}.$$

Only the upper bound is new; the lower bound is BFV's construction [5, Section 2]. By partial summation one obtains Problem #1190.

Corollary 1.2. *As* $m \to \infty$*,* $\varepsilon_m = L(-1+o(1), m) = \exp\bigl\{-(1+o(1))\sqrt{\log m \, \log_2 m}\bigr\}$.

The lower bound $\varepsilon_m \ge L(-1+o(1), m)$ follows from BFV's lower bound for f (as noted on [4]); the upper bound needs Theorem 1.1.

1.4. **What is new, and what is not.** The upper-bound argument is BFV's, step for step: the pruning of an extremal family [5, Section 4.1], the descending chain [5, Section 4.2], which they credit to Croot, and the final optimisation [5, Section 4.3]. BFV also showed [5, Theorem 2] that their conjecture would follow from a purely combinatorial "dense-core" statement about intersecting families [5, Conjecture 2]; see Remark 3.4. The only new point, due to the original note [20], is that a *weaker* dense-core statement, whose loss depends logarithmically on the size of the sets, is already enough for BFV's argument, and that this weaker statement is an immediate consequence of the spread lemma of Alweiss, Lovett, Wu and Zhang [1], in the sharpened forms of Rao [26], Tao [27] and Bell, Chueluecha and Warnke [2]. Later developments on the problem forum, including a computer-checked proof, are described in Section 8.

Organisation. Section 2 explains the argument informally. Section 3 states the spread lemma and deduces the dense-core lemma. Section 4 collects the number-theoretic input from BFV, Section 5 runs the descending chain, Section 6 carries out the final optimisation, and Section 7 proves Corollary 1.2.

2. Overview of the proof

From disjoint classes to intersecting set systems. Let $\mathcal{Q} \subseteq [1, x]$ be admissible with residues (a_q). Suppose that a subfamily $\mathcal{Q}^*$ of moduli all have the same exact divisor P (that is, $P \mid q$ and $\gcd(P, q/P) = 1$), and that their residues all agree modulo P. For $q \in \mathcal{Q}^*$ let $A(q)$ be the set of primes dividing q/P. If $A(q)$ and $A(q')$ were disjoint, then $\gcd(q, q')$ would divide P, so $a_q \equiv a_{q'} \pmod{\gcd(q, q')}$ and the two classes would meet. Hence $\{A(q) : q \in \mathcal{Q}^*\}$ is an *intersecting* family of sets of primes.

The descending chain. BFV, following Croot, exploit this repeatedly (Fig. 2). After some pruning, all moduli have exactly K distinct prime factors and distinct squarefree kernels. Starting from $P = 1$, at step r one

(i) finds a non-empty set C_r of primes contained in many of the supports $A(q)$, and keeps those moduli;
(ii) fixes the exact prime-power part P_r of q on C_r, keeping the most popular value (this costs a factor $h(P_r)^2$ up to $(\pi^2/6)^{|C_r|}$, where h is the product of the exponents);
(iii) fixes the residue modulo P_r, keeping the most popular one (this costs a factor P_r);

and replaces P by PP_r. After $R \le K$ steps every prime is fixed and a single modulus $P_1 \cdots P_R \ge x^{1-o(1)}$ survives. Meanwhile the number of survivors after step (ii) is at most the number of $m \le x/(P_1 \cdots P_r)$ with the right number of prime factors, which is small by a Rankin-type bound. Comparing the two counts bounds each P_r from above in terms of $S' = xL(-\sigma, x)$, the size of the pruned family; multiplying over r and using $\prod P_r \ge x^{1-o(1)}$ gives a lower bound for σ.

Where the loss is. Write $X = \log x$, $Y = \log_2 x$, and measure the number of prime factors and the number of steps on the scale $M = \sqrt{X/Y}$: $K = dM$ and $R = cM$ (we shall see $0 < c \le d \le 3$). Two of the losses in the chain are intrinsic: the Rankin-type count and the residue step together produce, after taking logarithms and dividing by $\log x$, the quadratic expression $c\sigma - c^2/4$, and the chain argument gives

$$1 \le c\sigma - \tfrac{c^2}{4} + (\text{loss from step (i)}) + o(1).$$

If step (i) cost nothing, then $1 \le c\sigma - c^2/4 \le \sigma^2$ would give $\sigma \ge 1$, which is the conjecture. The question is therefore what fraction of the moduli one can keep in step (i).

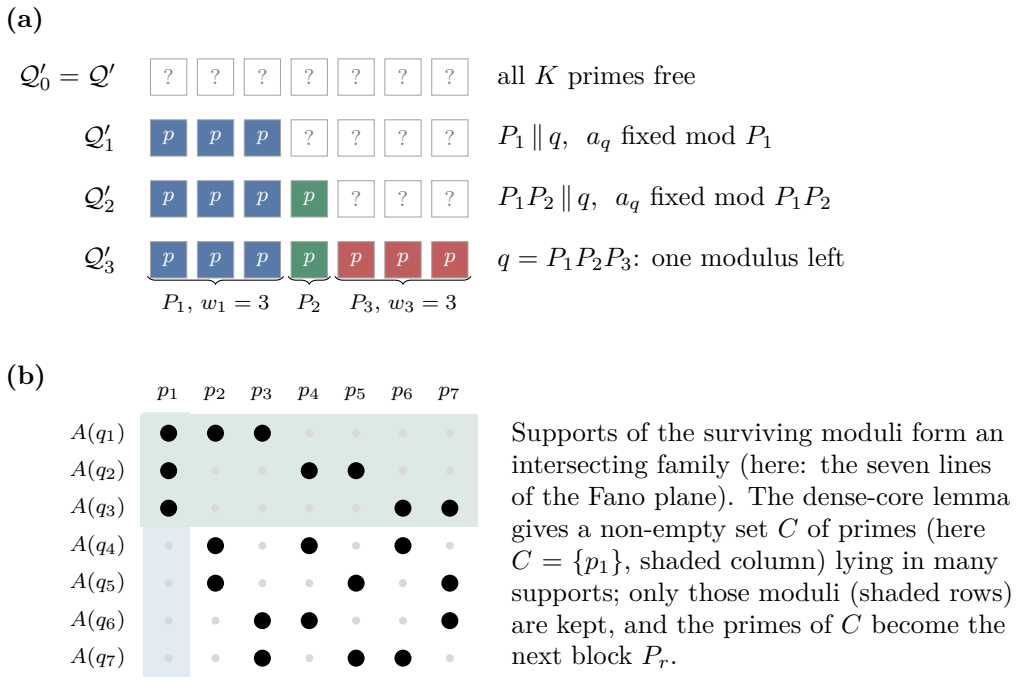


Figure 2. The descending chain of Section 5, schematically. (a) A surviving modulus at stages $r = 0, 1, 2, 3$ when $K = 7$; coloured cells are the primes already fixed (with their exact exponents) in the blocks P_1, P_2, P_3, and "?" marks primes that still vary over the surviving family. At stage r all residues a_q agree modulo $P_1 \cdots P_r$. (b) One step: how P_r is found.

BFV pass to a minimal intersecting family and use a bound in the spirit of Erdős and Lovász [16, p. 621]: a minimal intersecting family of sets of size at most K has at most wK^{w-1} members of size w [5, Lemma 3.4]. This finds a w-set C contained in at least a $4^{-w}K^{-(w-1)}$ proportion of the supports. Since $\log K \sim \frac{1}{2}\log_2 x$, this loss is of the same order as the main terms; it adds $\frac{c(d-c)}{2}$ to the right-hand side above, and after optimisation leads to the exponent $\sqrt{3}/2$ (Remark 6.1, Fig. 3).

The spread lemma (Theorem 3.2) says that a family of k-sets in which no set C is contained in more than a $\kappa^{-|C|}$ proportion of the members contains two disjoint members, as soon as $\kappa \geq C_0 \log(ek)$. Applied to an intersecting family, it produces a set C in at least a $(C_0 \log eK)^{-|C|}$ proportion of the supports (Corollary 3.3). The loss is now $(C_0 \log eK)^w$, i.e. $\exp(O(w \log\log K))$ rather than $\exp(\frac{1}{2}(w-1)\log_2 x)$, and its total contribution is $o(\log x)$.

3. Spread families and a dense core

For a finite family $\mathcal{F}$ of finite sets and a set T we write $\mathcal{F}_T := \{F \in \mathcal{F} : T \subseteq F\}$.

Definition 3.1 (Spread families)**.** Let $\kappa \geq 1$. A non-empty finite family $\mathcal{F}$ of finite sets is *κ-spread* if

$$|\mathcal{F}_T| \leq \kappa^{-|T|}\,|\mathcal{F}| \qquad \text{for every non-empty set } T.$$

This is the normalisation of Frankston, Kahn, Narayanan and Park [19, (4)] and of Tao [27]; Alweiss, Lovett, Wu and Zhang [1, Definition 1.10], who introduced the term, add the condition $|\mathcal{F}| \geq \kappa^k$ for k-uniform $\mathcal{F}$, which is automatic here (take $T \in \mathcal{F}$). Rao [26] and Bell–Chueluecha–Warnke [2] use a closely related "absolute" normalisation. The fundamental

fact about spread families is that they behave like random families, and in particular contain many disjoint members.

Theorem 3.2 (Spread lemma; Alweiss–Lovett–Wu–Zhang, Rao, Tao)**.** *There is an absolute constant C_1 with the following property. Let $r \geq 2$ and $k \geq 1$ be integers and let $\mathcal{F}$ be a κ-spread family of non-empty sets, each of size at most k, with $\kappa \geq C_1\, r \log(2rk)$. Then $\mathcal{F}$ contains r pairwise disjoint members.*

In exactly this normalisation the statement is Proposition 3 ("locating the petals") of Tao's exposition [27] (stated there for $k \geq 2$, which is no loss, since sets of size at most 1 also have size at most 2). It is the key step in the proofs of the improved sunflower bounds: Alweiss, Lovett, Wu and Zhang [1, Theorem 2.5 and Lemma 1.6] proved it with κ of order $r^2(\log k \log\log k + (\log r)^2)$, Rao [26, Lemma 2] with κ of order $r\log(rk)$, and Bell, Chueluecha and Warnke [2, Lemma 2] with κ of order $r\log k$. In each case the disjoint members are found by colouring the ground set at random ([1, Lemma 1.6]). We need only the case $r = 2$, for which all of these forms agree up to the constant.

Corollary 3.3 (Dense core)**.** *There is an absolute constant $C_0 \geq 1$ with the following property. Let $K \geq 1$, let $1 \leq k \leq K$, and let $\mathcal{A}$ be a non-empty intersecting family of distinct k-element sets. Then some non-empty set C satisfies*

$$|\mathcal{A}_C| > \frac{|\mathcal{A}|}{(C_0 \log(eK))^{|C|}}. \tag{1}$$

Proof. Let $C_0 := 4C_1$. If (1) failed for every non-empty C, then $\mathcal{A}$ would be κ-spread with $\kappa = C_0\log(eK) \geq 4C_1\log(ek) \geq 2C_1\log(4k)$ (the last inequality is $4k \leq e^2k^2$). By Theorem 3.2 with $r = 2$, $\mathcal{A}$ would contain two disjoint members, which are distinct since $k \geq 1$. This contradicts the assumption that $\mathcal{A}$ is intersecting. □

The deduction is the familiar "core or spread" dichotomy behind the sunflower bounds (compare [1, Lemma 2.4] and Lemma 2 of [27], which go back to the argument of Erdős and Rado [17]), and it underlies the spread approximation method of Kupavskii and Zakharov [21]. We have not found Corollary 3.3 stated in this form in the literature.

Remark 3.4 (Relation to BFV's Conjecture 2). BFV conjectured [5, Conjecture 2] that there are constants $t(1), t(2), \ldots$ with $\log t(j) = o(j\log j)$ such that every finite intersecting family $\mathcal{A}$ has a non-empty C with $|\mathcal{A}_C| \geq |\mathcal{A}|/t(|C|)$, and proved [5, Theorem 2] that this implies $f(x) = xL(-1+o(1), x)$. They remarked that the conclusion holds with $t(j) \ll j^{j+2}$, which is just too weak. Corollary 3.3 supplies $t(j) = (C_0 \log eK)^j$ for families of sets of size at most K. This is not Conjecture 2, because it depends on K. It is nonetheless enough: BFV's proof of their Theorem 2 only uses the bound $\sum_j \log t(w_j) = o(\sqrt{\log x \log_2 x})$ for block sizes with $\sum_j w_j = K \ll \sqrt{\log x/\log_2 x}$, and here $\sum_j \log t(w_j) = K\log(C_0 \log eK) = O(K\log\log K)$. The argument below is a self-contained version of this.

Remark 3.5 (Some dependence on K is necessary). Some growth in K is unavoidable in a bound of the form $(\text{const})^{|C|}$: for every integer $s \geq 2$ there is an intersecting, $(s/2)$-spread family of k-sets with $k = s^s + s - 1$, so that $s/2 \sim \frac{1}{2}\log k/\log\log k$. Take $t = s^s$ disjoint blocks $B_1, \ldots, B_t$ of size s, and let $\mathcal{A}$ consist of the sets $B_i \cup \{x_l : l \neq i\}$ with $x_l \in B_l$ (one whole block plus one point from each other block). Two members meet, because each contains a point of the other's whole block. If C meets m blocks, each in one point, then $|\mathcal{A}|/|\mathcal{A}_C| = ts^m/(ms + t - m) \geq (s/2)^m$; if C meets one block in $c \geq 2$ points and $m-1$ others in one point each, then $|\mathcal{A}|/|\mathcal{A}_C| = ts^{m-1} \geq (s/2)^{c+m-1}$ because $t = s^s \geq (s/2)^c$; otherwise $\mathcal{A}_C = \emptyset$. (A brute-force check for small s is in `figdata.py`.)

Remark 3.6 (Other routes). Theorem 3.2 with $r = 2$ also follows from the theorem of Frankston, Kahn, Narayanan and Park [19], and from the Park–Pham theorem [25] (the Kahn–Kalai conjecture); the original note [20] took the latter route. In outline: if $\mathcal{F}$ is κ-spread and $\langle\mathcal{F}\rangle$ denotes its up-closure, any cover $\mathcal{G}$ of $\langle\mathcal{F}\rangle$ satisfies $|\mathcal{F}| \le \sum_{G\in\mathcal{G}} |\mathcal{F}_G| \le |\mathcal{F}| \sum_G \kappa^{-|G|}$, so the expectation threshold is at most κ^{-1}; Park–Pham then gives $p_c(\langle\mathcal{F}\rangle) \le K_{\mathrm{PP}}\kappa^{-1}\log\max(k,2)$. If this is at most $1/(2r)$, colour the ground set randomly with $2r$ colours: each colour class contains a member of $\mathcal{F}$ with probability at least $1/2$, so some colouring has r such classes, giving r disjoint members. (The use of $2r$ colours and linearity of expectation, rather than r colours and a union bound, is the device of [2].)

4. Input from de la Bretèche, Ford and Vandehey

Notation. From now on x is large and

$$X := \log x, \qquad Y := \log_2 x, \qquad M := \sqrt{X/Y}, \qquad Z := \sqrt{XY},$$

so that $L(\alpha, x) = e^{\alpha Z}$, $Z = MY$ and $X = MZ$. We write $\omega(n)$ for the number of distinct prime factors of n, $\mathrm{rad}(n)$ for its squarefree kernel, $v_p(n)$ for the exponent of p in n, and, following BFV,

$$h(n) := \prod_{p^\nu \,\|\, n} \nu.$$

We write $D \,\|\, n$ ("D is an exact divisor of n") if $D \mid n$ and $\gcd(D, n/D) = 1$; then $h(n) = h(D)\,h(n/D) \ge h(D)$. Every $o(1)$ denotes a function of x alone that tends to 0 as $x \to \infty$; in particular it is uniform in all other parameters.

Lemma 4.1 (Chinese remainder criterion)**.** *The classes $a \pmod q$ and $b \pmod{q'}$ have a common element if and only if $a \equiv b \pmod{\gcd(q, q')}$.*

Proof. A common element n gives $a \equiv n \equiv b \pmod{\gcd(q,q')}$. Conversely, if $\gcd(q,q') \mid a-b$, write $a - b = uq + u'q'$ (Bézout); then $n = a - uq = b + u'q'$ is common to both classes. □

The first estimate is a Rankin-type bound.

Lemma 4.2 ([5, Lemma 3.1])**.** *Fix $A > 0$. Uniformly for $2 \le y \le x$ and $0 \le \alpha \le A$,*

$$\#\{n \le y : \omega(n) \ge \alpha M\} \le yL(-\alpha/2 + o(1), x).$$

(The proof bounds the count by $z^{-\alpha M}\sum_{n\le y} z^{\omega(n)}$ with $z = \sqrt{X}/Y$.) We use it in the following form.

Lemma 4.3. *There is a function $\eta(x) \ge 0$ tending to 0 such that, for all integers $0 \le W \le K \le 3M$ and all $1 \le y \le x$,*

$$\#\{n \le y : \omega(n) = K - W\} \le y\,L\Big(-\frac{K}{2M} + \eta(x), x\Big)(\log x)^{W/2}.$$

Proof. Let $\eta(x) \ge 0$ be the $o(1)$ of Lemma 4.2 with $A = 3$ (made non-negative). If $W = K$ the left side is $\#\{n \le y : n = 1\} \le 1 \le y$, while the right side is $y\,e^{\eta Z} \ge y$, since $(\log x)^{K/2} = e^{KY/2} = L(K/(2M), x)$. If $W < K$ and $y < 2$ the left side is 0. If $W < K$ and $y \ge 2$, apply Lemma 4.2 with $\alpha = (K-W)/M \in (0, 3]$ to the larger set $\{n \le y : \omega(n) \ge K - W\}$ and note $L(W/(2M), x) = e^{WZ/(2M)} = e^{WY/2} = (\log x)^{W/2}$. □

The second input is BFV's pruning of an extremal family. Fix an admissible $\mathcal{Q} \subseteq [1, x]$ with $|\mathcal{Q}| = f(x) =: S$, and residues $(a_q)_{q\in\mathcal{Q}}$.

Proposition 4.4 (Pruning; [5, Section 4.1])**.** *There are a subfamily $\mathcal{Q}' \subseteq \mathcal{Q}$ and an integer K such that, with $S' := |\mathcal{Q}'|$,*

(P1) *$S' \geq S\,L(-o(1), x)$;*
(P2) *$xL(-2,x) \leq q \leq x$ for every $q \in \mathcal{Q}'$;*
(P3) *$h(q) \leq e^{\sqrt{X}}$ for every $q \in \mathcal{Q}'$;*
(P4) *$\omega(q) = K$ for every $q \in \mathcal{Q}'$, and $1 \leq K \leq 3M$;*
(P5) *the kernels $\mathrm{rad}(q)$, $q \in \mathcal{Q}'$, are distinct.*

Proof sketch. These are conditions (1)–(5) of [5, Section 4.1] (where rad is called ker); we recall why each costs little. By BFV's lower bound, $S \geq xL(-1-o(1),x)$. Discarding $q < xL(-2,x)$ removes $o(S)$ moduli. By Lemma 4.2 with $\alpha = 3$, at most $xL(-\frac{3}{2}+o(1),x) = o(S)$ moduli have $\omega(q) \geq 3M$. By [5, Lemma 3.2], at most $x\exp(-\frac{1}{5}\sqrt{X}\,Y) = o(S)$ integers $n \leq x$ have $h(n) \geq e^{\sqrt{X}}$ (note $\sqrt{X}\,Y/Z = \sqrt{Y} \to \infty$). Pigeonholing the value of $\omega(q)$ costs a factor at most $3M = L(o(1),x)$. Finally, [5, Lemma 3.3] shows that at most $H^2(\pi^2/6)^K \leq H^2 2^K$ integers n with $h(n) \leq H$ share a given squarefree kernel with K prime factors; with $H = e^{\sqrt{X}}$ this is $e^{2\sqrt{X}}2^{3M} = L(o(1),x)$, and keeping one modulus per kernel gives (P5). Since $xL(-2,x) > 1$, (P2) forces $K \geq 1$. □

5. The descending chain

Fix $\mathcal{Q}'$, K and S' as in Proposition 4.4, put $d := K/M \in (0,3]$, let C_0 be the constant of Corollary 3.3, and set

$$\Lambda := \frac{\pi^2}{6}\,C_0 \log(eK).$$

The following is BFV's descending chain [5, Section 4.2], with the dense-core step changed.

Proposition 5.1 (Chain inequality)**.** *There are an integer $1 \leq R \leq K$ and pairwise coprime integers $P_1, \dots, P_R > 1$ such that, writing $P_{\leq r} := P_1 \cdots P_r$ and $W_r := \omega(P_{\leq r})$,*

(a) *$W_R = K$ and $P_{\leq R} \in \mathcal{Q}'$; in particular $P_{\leq R} \geq xL(-2,x)$;*
(b) *for every $1 \leq r \leq R$,*

$$P_r \leq \frac{x}{S'}\,L\Big(-\frac{d}{2}+\eta(x),x\Big)(\log x)^{W_r/2}\,\Lambda^{W_r}\,e^{2\sqrt{X}}, \tag{2}$$

with η as in Lemma 4.3.

Proof. Construction. We construct $P_1, P_2, \dots$ and subfamilies $\mathcal{Q}' = \mathcal{Q}'_0 \supseteq \mathcal{Q}_1 \supseteq \mathcal{Q}'_1 \supseteq \mathcal{Q}_2 \supseteq \cdots$ inductively. Write $P_{<r} := P_{\leq r-1}$ (so $P_{<1} = 1$). Suppose that $r \geq 1$ and that a non-empty $\mathcal{Q}'_{r-1} \subseteq \mathcal{Q}'$ has been found such that

$$P_{<r} \,\|\, q \ \text{ for all } q \in \mathcal{Q}'_{r-1}, \qquad \text{and all } a_q\ (q \in \mathcal{Q}'_{r-1}) \text{ agree modulo } P_{<r}. \tag{3}$$

This holds for $r = 1$. If $W_{r-1} = K$ we stop and set $R := r-1$. Otherwise, for $q \in \mathcal{Q}'_{r-1}$ let $A(q)$ be the set of primes dividing $q/P_{<r}$, i.e. the primes of q not dividing $P_{<r}$; then $|A(q)| = K - W_{r-1} =: k$, and $1 \leq k \leq K$.

The supports are distinct: if $A(q_1) = A(q_2)$ then $\mathrm{rad}(q_1) = \mathrm{rad}(P_{<r})\prod_{p\in A(q_1)} p = \mathrm{rad}(q_2)$, so $q_1 = q_2$ by (P5).

The supports are intersecting: if $A(q_1) \cap A(q_2) = \emptyset$ for some $q_1 \neq q_2$, then every prime dividing $\gcd(q_1,q_2)$ divides $P_{<r}$, and for such p we have $v_p(q_1) = v_p(P_{<r}) = v_p(q_2)$ by (3); hence $\gcd(q_1,q_2) \mid P_{<r}$. As $a_{q_1} \equiv a_{q_2} \pmod{P_{<r}}$, Lemma 4.1 shows that the two classes meet, contradicting admissibility.

Step (i): dense core. By Corollary 3.3 applied to $\{A(q) : q \in \mathcal{Q}'_{r-1}\}$, there is a non-empty set C_r of primes, of size $w_r := |C_r|$, such that

$$\mathcal{S}_r := \{q \in \mathcal{Q}'_{r-1} : C_r \subseteq A(q)\} \quad \text{satisfies} \quad |\mathcal{S}_r| > \frac{|\mathcal{Q}'_{r-1}|}{(C_0 \log eK)^{w_r}}.$$

Step (ii): exact powers. For $q \in \mathcal{S}_r$ let $B(q) := \prod_{p \in C_r} p^{v_p(q)}$, and for each value u of B let $N_u := \#\{q \in \mathcal{S}_r : B(q) = u\}$. Every value has the form $u = \prod_{p \in C_r} p^{\nu_p}$ with all $\nu_p \geq 1$, and $h(u) = \prod \nu_p$, so

$$\sum_u h(u)^{-2} \leq \prod_{p \in C_r} \sum_{\nu \geq 1} \nu^{-2} = \Big(\frac{\pi^2}{6}\Big)^{w_r}.$$

Since $\sum_u N_u = |\mathcal{S}_r|$, some value $u =: P_r$ has $N_{P_r} \geq |\mathcal{S}_r|\, h(P_r)^{-2} (\pi^2/6)^{-w_r}$ (otherwise summing over u gives $|\mathcal{S}_r| < |\mathcal{S}_r|$). Put $\mathcal{Q}_r := \{q \in \mathcal{S}_r : B(q) = P_r\}$. Then

$$|\mathcal{Q}_r| > \frac{|\mathcal{Q}'_{r-1}|}{\Lambda^{w_r}\, h(P_r)^2}. \tag{4}$$

Step (iii): residues. Let $\mathcal{Q}'_r$ be the set of $q \in \mathcal{Q}_r$ whose residue a_q mod P_r is the most frequent one; then

$$|\mathcal{Q}'_r| \geq |\mathcal{Q}_r|/P_r. \tag{5}$$

The invariant is preserved. The primes of P_r lie in C_r, hence do not divide $P_{<r}$, so $P_r > 1$ is coprime to $P_{<r}$. For $q \in \mathcal{Q}'_r$ we have $P_{<r} \,\|\, q$ and $v_p(q) = v_p(P_r)$ for $p \in C_r$, so $P_{\leq r} = P_{<r} P_r \,\|\, q$. The residues a_q, $q \in \mathcal{Q}'_r$, agree modulo $P_{<r}$ and modulo P_r, hence modulo $P_{\leq r}$. Moreover $\mathcal{Q}'_r \neq \emptyset$ and $W_r = W_{r-1} + w_r > W_{r-1}$, so the process stops after $R \leq K$ steps, and $R \geq 1$ because $W_0 = 0 < K$.

Part (a). When the process stops, every $q \in \mathcal{Q}'_R$ satisfies $P_{\leq R} \,\|\, q$ and $\omega(q) = K = \omega(P_{\leq R})$, so $q/P_{\leq R}$ has no prime factor and $q = P_{\leq R}$. Thus $\mathcal{Q}'_R = \{P_{\leq R}\}$ and $P_{\leq R} \in \mathcal{Q}'$, and (P2) gives $P_{\leq R} \geq xL(-2, x)$.

Part (b). Fix $1 \leq r \leq R$. Iterating (4) and (5), and using $h(P_{\leq r}) = \prod_{j \leq r} h(P_j)$ (the P_j are pairwise coprime) and $W_r = \sum_{j \leq r} w_j$,

$$|\mathcal{Q}_r| > S' \prod_{j<r} \frac{1}{P_j} \prod_{j \leq r} \frac{1}{\Lambda^{w_j} h(P_j)^2} = \frac{S'}{P_{<r}\, \Lambda^{W_r}\, h(P_{\leq r})^2}. \tag{6}$$

On the other hand, each $q \in \mathcal{Q}_r$ is $q = P_{\leq r} m$ with $m \leq x/P_{\leq r}$ and $\omega(m) = K - W_r$, and distinct q give distinct m. Since $1 \leq m \leq x/P_{\leq r} \leq x$, Lemma 4.3 gives

$$|\mathcal{Q}_r| \leq \frac{x}{P_{\leq r}}\, L\Big(-\frac{d}{2} + \eta(x), x\Big)(\log x)^{W_r/2}. \tag{7}$$

Since $P_{\leq r} \,\|\, q$ for $q \in \mathcal{Q}_r$, (P3) gives $h(P_{\leq r}) \leq h(q) \leq e^{\sqrt{X}}$. Combining (6) and (7) and cancelling $P_{<r}$ gives (2). □

Remark 5.2. In BFV's version [5, (4.1)] the factor Λ^{w_r} in (4) is $7^{w_r} K^{w_r - 1}$. Every other step is the same.

6. Proof of Theorem 1.1

Proof of Theorem 1.1. The lower bound is [5, Section 2]. For the upper bound, take $\mathcal{Q}'$, S', K from Proposition 4.4 and R, $P_1, \dots, P_R$ from Proposition 5.1. Define $\sigma = \sigma(x)$ by $S' = xL(-\sigma, x)$; then $\sigma \geq 0$ because $S' \leq x$. Put $c := R/M$ and $d := K/M$, so $0 < c \leq d \leq 3$.

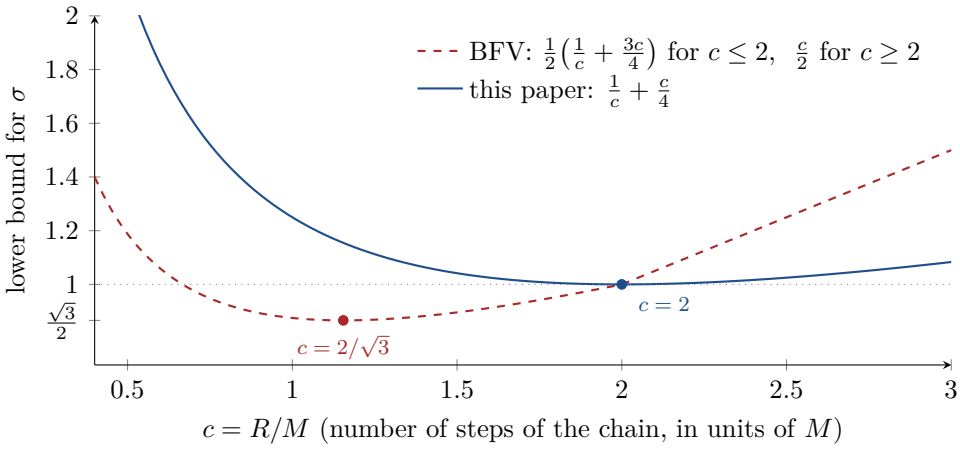


Figure 3. The final optimisation. The extremal family has $|\mathcal{Q}'| = xL(-\sigma, x)$, and the chain has $R = cM$ steps; neither c nor $d = K/M$ can be chosen, so the guaranteed bound is the minimum over c of the curve. Solid: inequality (8) gives $\sigma \ge \frac{1}{c} + \frac{c}{4}$, with minimum 1 at $c = 2$. Dashed: in BFV's argument the core step adds $\frac{c(d-c)}{2}$, giving $\sigma \ge \frac{1}{c} + \frac{3c}{4} - \frac{d}{2}$; together with $\sigma \ge \frac{d}{2}$ (from Lemma 4.2 applied to $\mathcal{Q}'$ itself) and the worst admissible $d \in [c, 3]$, this is the dashed curve, with minimum $\sqrt{3}/2$ at $c = 2/\sqrt{3}$ (Remark 6.1).

Multiply (2) over $1 \le r \le R$ and use (a):

$$xL(-2,x) \le P_{\le R} \le \Big(\frac{x}{S'} L\Big(-\frac{d}{2}+\eta, x\Big)\Big)^R (\log x)^{T/2}\, \Lambda^T\, e^{2R\sqrt{X}}, \qquad T := \sum_{r=1}^{R} W_r.$$

Since each later block contains at least one prime, $W_r \le K - (R - r)$, so

$$T \le \sum_{r=1}^{R} (K - R + r) = RK - \frac{R^2}{2} + \frac{R}{2}.$$

Take logarithms, recalling $\log(x/S') = \sigma Z$, $RZ = cX$, $RKY = cdX$ and $R^2 Y = c^2 X$:

$$X - 2Z \le c\sigma X - \frac{cd}{2}X + c\eta X + \frac{cd}{2}X - \frac{c^2}{4}X + \frac{RY}{4} + T\log\Lambda + 2R\sqrt{X}.$$

The terms $\pm\frac{cd}{2}X$ cancel. Dividing by X,

$$1 - \varepsilon(x) \le c\sigma - \frac{c^2}{4}, \qquad \varepsilon(x) := \frac{2Z + 3\eta X + \frac{3}{4}MY + 9M^2\log\Lambda + 6M\sqrt{X}}{X}, \tag{8}$$

where we used $c \le 3$, $R \le 3M$ and $T \le RK \le 9M^2$. Each term of $\varepsilon(x)$ is $o(1)$, uniformly in all parameters: $Z/X = MY/X = \sqrt{Y/X} \to 0$; $\eta \to 0$; $M^2 \log\Lambda/X = \log\Lambda/Y$, where $\log\Lambda = \log\log(eK) + O(1) = O(\log Y)$ because $K \le 3M$; and $M\sqrt{X}/X = 1/\sqrt{Y} \to 0$. Now

$$c\sigma - \frac{c^2}{4} = \sigma^2 - \Big(\sigma - \frac{c}{2}\Big)^2 \le \sigma^2,$$

so (8) gives $\sigma \ge \sqrt{1 - \varepsilon(x)} = 1 - o(1)$, i.e. $S' \le xL(-1 + o(1), x)$. Finally (P1) gives $f(x) = S \le S' L(o(1), x) \le xL(-1 + o(1), x)$. □

Remark 6.1 (Where $\sqrt{3}/2$ came from). With BFV's core step, (4) has $7^{w_r}K^{w_r-1}$ in place of Λ^{w_r}, so the product over r acquires the extra factor $K^{\sum_r (W_r-r)}$, and $\sum_r (W_r - r) \le R(K-R)$ by $W_r \le K-(R-r)$. Since $\log K = (\frac{1}{2}+o(1))Y$, this adds $\frac{c(d-c)}{2}$ to the right side of (8), giving $1 \le c\sigma + \frac{cd}{2} - \frac{3c^2}{4} + o(1)$, i.e. $\sigma \ge \frac{1}{c} + \frac{3c}{4} - \frac{d}{2} - o(1)$, which is the bound $S' \le xL(-\frac{1}{c} - \frac{3c}{4} + \frac{d}{2} + o(1), x)$ of [5, Section 4.3]. Because d now appears, BFV combine it with $S' \le xL(-\frac{d}{2} + o(1), x)$, which holds because every $q \in \mathcal{Q}'$ has $\omega(q) = K = dM$. The worst case $d = \frac{1}{c} + \frac{3c}{4}$ gives $\sigma \ge \frac{1}{2}(\frac{1}{c} + \frac{3c}{4}) \ge \frac{\sqrt{3}}{2}$. In the argument above, d cancels and the second bound is not needed.

Remark 6.2 (Consistency with the construction). The argument can be close to sharp only when $c\sigma - c^2/4 \approx \sigma^2$, i.e. $c \approx 2\sigma$; for $\sigma \approx 1$ the chain must then have about $2M$ steps. BFV's lower-bound construction [5, Section 2] has exactly this shape. There each modulus is a product $p_0 p_1 \cdots p_r$ of $r+1$ primes with $r \sim 2M$: the prime p_0 is common to all moduli, and the residue of a_q modulo p_{k-1} determines p_k. Run on this family, the chain can fix one prime per step, starting with p_0, giving $R = K = r + 1 \sim 2M$.

7. Proof of Corollary 1.2

We need one elementary integral estimate.

Lemma 7.1. *For every fixed $\beta > 0$, as $m \to \infty$,*

$$\int_m^\infty \frac{dt}{t\, L(\beta, t)} \le L(-\beta + o(1), m).$$

Proof. Substitute $u = \log t$ and let $g(u) = \sqrt{u \log u}$, which is increasing for $u \ge 1$; with $u_0 = \log m$ the integral is $\int_{u_0}^\infty e^{-\beta g(u)}\, du$. Split $[u_0, \infty)$ into the intervals $[2^j u_0, 2^{j+1} u_0]$, $j \ge 0$. On the j-th interval $g(u) \ge g(2^j u_0) \ge 2^{j/2} g(u_0) \ge (1 + (\sqrt{2}-1)j)\, g(u_0)$, the last step by convexity of $j \mapsto 2^{j/2}$. Hence

$$\int_{u_0}^\infty e^{-\beta g(u)}\, du \le u_0 e^{-\beta g(u_0)} \sum_{j \ge 0} \big(2e^{-\beta(\sqrt{2}-1) g(u_0)}\big)^j \le 2u_0 e^{-\beta g(u_0)}$$

once $2e^{-\beta(\sqrt{2}-1)g(u_0)} \le \frac{1}{2}$. Since $\log(2u_0) = o(g(u_0))$ and $g(u_0) = \sqrt{\log m \log_2 m}$, the right side is $L(-\beta + o(1), m)$. □

Proof of Corollary 1.2. Upper bound. Fix $\delta \in (0,1)$. By Theorem 1.1 there is t_0 with $f(t) \le tL(-1+\delta, t)$ for $t \ge t_0$. Let $m \ge t_0$ and let $m < n_1 < \cdots < n_k$ carry pairwise disjoint classes. Put $A(t) := \#\{i : n_i \le t\}$. The moduli $n_i \le t$ form an admissible subset of $[1, t]$, so $A(t) \le f(t)$, and $A(t) = 0$ for $t < n_1$. By partial summation,

$$\sum_{i=1}^k \frac{1}{n_i} = \sum_{i=1}^k \int_{n_i}^\infty \frac{dt}{t^2} = \int_m^\infty \frac{A(t)}{t^2}\, dt \le \int_m^\infty \frac{dt}{t\, L(1-\delta, t)} \le L(-1+\delta+o(1), m)$$

by Lemma 7.1. Hence $\varepsilon_m \le L(-1+\delta+o(1), m)$ for every $\delta > 0$, i.e. $\varepsilon_m \le L(-1+o(1), m)$.

Lower bound. Let $x := mL(2, m)$. Then $\log x = (1+o(1)) \log m$ and $\log_2 x = \log_2 m + o(1)$, so $L(\alpha, x) = L(\alpha + o(1), m)$ for each fixed α. By BFV's lower bound there is an admissible set of $f(x) = xL(-1+o(1), x) = mL(1+o(1), m)$ moduli in $[1, x]$. At most m of them are $\le m$, and $m = o(f(x))$; the remaining $f(x)(1-o(1))$ moduli lie in $(m, x]$, so

$$\varepsilon_m \ge \frac{f(x)(1-o(1))}{x} = L(-1+o(1), x) = L(-1+o(1), m).$$ □

8. Subsequent work

The original note [20] was announced on the forum thread for Problem #202 [28] on 23 April 2026. Thomas Bloom replied that he would update the problem page once the proof had been formalised or checked by a human expert. On 14 May 2026 the forum user *paws* posted a Lean 4/Mathlib formalisation of Theorem 1.1 and Corollary 1.2 [22], depending only on Lean's core axioms; Nat Sothanaphan confirmed it, noting that the Park–Pham input used there was itself formalised rather than assumed. Both problem pages now record the result [3, 4].

On the thread for Problem #1190 [29], MalekZ posted on 28 April 2026 a short note [23] deriving $\varepsilon_m = L(m)^{-1+o(1)}$ from the asymptotic for f by Abel summation and truncation of an extremal family, as in Section 7, and on 30 April 2026 Przemek Chojecki posted a separate write-up produced with GPT-5.5 Pro [7], which proves Theorem 1.1 by essentially the same route (BFV's chain together with the spread lemma), as Sothanaphan observed. Exact values of $f(n)$ for $n \le 100$ are in OEIS A389975 [24].

Acknowledgements and provenance

The results were found by GPT-5.4 Pro, prompted by Boon Suan Ho, and first written up in [20]. This paper uses comments by Thomas Bloom, paws and Nat Sothanaphan on the forum thread https://www.erdosproblems.com/forum/thread/202, and by MalekZ, Przemek Chojecki and Nat Sothanaphan on https://www.erdosproblems.com/forum/thread/1190. This text was written by Claude Opus 5.5.

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